# Chemistry

# Class 9 (English Medium)

# Solved Exercises — Sindh Textbook Board

8 lessons — 376 questions

# علم کیمیا جا بنیاد1باب :

Question: Which branch deals with hydrocarbon compounds? (a) Industrial Chemistry (b) Inorganic Chemistry (c) Organic Chemistry (d) Physical Chemistry

Answer: Correct answer: (c) Organic Chemistry, because this branch studies carbon and hydrogen compounds.

Question: What is the atomic mass of an element expressed in grams called? (a) Gram molecular mass (b) Gram atomic mass (c) Gram formula mass (d) Mole

Answer: Correct answer: (b) Gram atomic mass, i.e. the mass of one mole of atoms in grams.

Question: Which of the following can be separated by a physical method? (a) Mixture (b) Element (c) Compound (d) Substance

Answer: Correct answer: (a) Mixture, because its components can be separated by physical methods such as filtration, evaporation and distillation.

Question: What is the molar mass of H2SO4? (a) 98 a.m.u (b) 9.8 gram (c) 98 gram (d) 9.8 a.m.u

Answer: Correct answer: (c) 98 gram, because the molecular mass of H2SO4 is 98 a.m.u, so one mole weighs 98 gram.

Question: What is a molecule containing two atoms called? (a) Monoatomic molecule (b) Diatomic molecule (c) Heteroatomic molecule (d) Polyatomic molecule

Answer: Correct answer: (b) Diatomic molecule, such as H2, O2 and Cl2.

Question: Which formula shows the actual number of atoms in a molecule? (a) Chemical formula (b) Empirical formula (c) Molecular formula (d)

Formula mass

Answer: Correct answer: (c) Molecular formula, such as C6H12O6 for glucose.

Question: Who prepared ethyl alcohol? (a) Ibn Sina (b) Al-Razi (c) Al-Beruni (d) Jabir bin Hayyan

Answer: Correct answer: (b) Al-Razi, who prepared ethyl alcohol by the fermentation method.

Question: Which of the following is a homoatomic molecule? (a) H2 (b) NH3 (c) H2O (d) CO2

Answer: Correct answer: (a) H2, because both its atoms belong to the same element.

Question: What is the empirical formula of hydrogen peroxide? (a) H2O2 (b) HO (c) OH2 (d) O2H2

Answer: Correct answer: (b) HO, because the simplest ratio of H to O in H2O2 is 1:1.

Question: What is the pure form of matter called? (a) Radical (b) Mixture (c) Compound (d) Substance

Answer: Correct answer: (d) Substance, i.e. the pure form of matter with fixed composition and fixed properties.

Question: What is the difference between physical chemistry and analytical chemistry?

Answer: Physical chemistry studies the physical properties, changes and energy of matter, while analytical chemistry deals with identifying the composition of matter and measuring the amounts of its components.

Question: Write the types of molecules according to the number of atoms.

Answer: Monoatomic molecule: He, Ne. Diatomic: H2, O2. Triatomic: H2O, CO2. Polyatomic: H2SO4, C6H12O6.

Question: Differentiate between: (a) atom and ion (b) molecule and

molecular ion (c) compound and mixture

Answer: (a) An atom is neutral; an ion carries a positive or negative charge. (b) A molecule is a neutral group of atoms; a molecular ion is a charged group such as NH4+ and SO4--. (c) In a compound the components are joined by chemical bonds in a fixed ratio; in a mixture the components are physically mixed in any ratio.

Question: Define the following terms: (a) Gram atomic mass (b) Gram molecular mass (c) Gram formula mass

Answer: (a) Gram atomic mass: the mass of one mole of atoms of an element in grams. (b) Gram molecular mass: the mass of one mole of molecules of a substance in grams. (c) Gram formula mass: the mass of one mole of formula units of an ionic compound in grams.

Question: Write the chemical, empirical and molecular formulas of: sulphuric acid, carbon dioxide, glucose, benzene

Answer: Sulphuric acid: chemical H2SO4, empirical H2SO4, molecular H2SO4. Carbon dioxide: all three CO2. Glucose: chemical C6H12O6, empirical CH2O, molecular C6H12O6. Benzene: chemical C6H6, empirical CH, molecular C6H6.

Question: What is a free radical?

Answer: A free radical is an atom or group having one unpaired electron, such as Cl• and CH3•. It is very reactive.

Question: What is the relation between the empirical formula and the molecular formula?

Answer: Molecular formula = n × empirical formula, where n = molecular mass ÷ empirical formula mass. Example: the empirical formula of glucose is CH2O and n = 6, so its molecular formula is C6H12O6.

Question: Hydrogen and oxygen are considered elements, but water is not. Why? Explain.

Answer: Hydrogen and oxygen each consist of only one kind of atom, so they are elements. Water is a compound of hydrogen and oxygen joined chemically in a fixed ratio, and its properties are quite different from those of its components.

Question: What is meant by chemical species? Explain atom, molecule, ion and free radical.

Answer: A chemical species is a basic particle taking part in a chemical process. Atom: the smallest particle of an element. Molecule: a neutral group of two or more atoms. Ion: a charged atom or group. Free radical: a particle with an unpaired electron.

Question: What are the uses of chemistry in daily life?

Answer: Chemistry serves every field of daily life: cooking and preserving food, medicines, cleaning products, fertilizers and pesticides in agriculture, fuels, clothes, plastics and paints.

Question: Explain the empirical formula and the molecular formula in detail.

Answer: The empirical formula shows the simplest whole-number ratio of atoms in a compound, such as CH2O for glucose. The molecular formula shows the actual number of atoms in a molecule, such as C6H12O6 for glucose. Molecular formula = n × empirical formula, where n is a whole number.

Question: State the steps of balancing equations.

Answer: First: write the unbalanced equation. Second: count the atoms of each element on both sides. Third: equalise the numbers of atoms with coefficients, but never change the subscripts. Fourth: re-count at the end to check.

Question: Give the names of the branches of chemistry and explain five branches.

Answer: Main branches of chemistry: (1) Physical chemistry: study of the physical properties of matter. (2) Organic chemistry: study of carbon compounds. (3) Inorganic chemistry: study of compounds other than carbon. (4) Analytical chemistry: study of the composition and amounts of matter. (5) Biochemistry: study of chemical processes in living things.

Question: Check and balance the following equations: (a) $NH_3 + O_2 \rightarrow NO + H_2O$ (b) $KNO_3 \rightarrow KNO_2 + O_2$ (c) $Ca + H_2O \rightarrow Ca(OH)_2 + H_2$ (d) $NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$ (e) $CO + O_2 \rightarrow CO_2$

Answer: (a) $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$. (b) $2KNO_3 \rightarrow 2KNO_2 + O_2$. (c) $Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$. (d) $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$. (e) $2CO + O_2 \rightarrow 2CO_2$.

Question: Find the formula mass (a.m.u) of: $Al_2O_3$, $MgCl_2$, NaCl, $KNO_3$

Answer: $Al_2O_3$ =

= $2\times27 + 3\times16$

= 102 a.m.u.$MgCl_2$ =

= $24 + 2\times35.5$

= 95 a.m.u.NaCl =

= $23 + 35.5$

= 58.5 a.m.u.$KNO_3$ =

= $39 + 14 + 3\times16$

= 101 a.m.u.

Question: Find the molecular mass (a.m.u) of: C2H5OH, H2O, NH3, CO2

Answer: C2H5OH =

= 2×12 + 6×1 + 16

= 46 a.m.u.H2O =

= 2×1 + 16

= 18 a.m.u.NH3 =

= 14 + 3×1

= 17 a.m.u.CO2 =

= 12 + 2×16

= 44 a.m.u.

Question: How many moles are needed to get 40 g of H2SO4?

Answer: Molar mass of H2SO4 = 98 g.Moles =

= given mass ÷ molar mass

= 40 ÷ 98

= 0.41 mol.

Question: Calculate the number of moles and the number of molecules in:

(a) 16 g of $H_2CO_3$ (b) 20 g of $C_6H_{12}O_6$

Answer: (a) Molar mass of $H_2CO_3$ = 62 g, moles = 16 ÷ 62 = 0.26 mol, molecules = 0.26 × $6.02\times10^{23}$ = $1.56\times10^{23}$. (b) Molar mass of $C_6H_{12}O_6$ = 180 g, moles = 20 ÷ 180 = 0.11 mol, molecules = 0.11 × $6.02\times10^{23}$ = $6.7\times10^{22}$.

Question: In-text question: How can you distinguish between matter and a chemical substance?

Answer: Matter is anything that has mass and volume. A chemical substance is the pure form of matter with fixed composition and properties, such as an element or a compound.

Question: In-text question: Which elements are present in the following compounds? Washing soda, sugar, sand, caustic soda

Answer: Washing soda ($Na_2CO_3$): sodium, carbon, oxygen. Sugar ($C_{12}H_{22}O_{11}$): carbon, hydrogen, oxygen. Sand ($SiO_2$): silicon, oxygen. Caustic soda (NaOH): sodium, oxygen, hydrogen.

Question: In-text question: Identify the element, compound or mixture in: table salt, ice cream, blood, silicon, Coca-Cola, tin, zinc, sulphur in water

Answer: Table salt: compound. Ice cream: mixture. Blood: mixture. Silicon: element. Coca-Cola: mixture. Tin: element. Zinc: element. Sulphur in water: mixture.

Question: In-text question: In the equation below, place the coefficient 4 before $KClO_3$ on the reactant side and 4 before KCl on the product side and balance it: $KClO_3 \rightarrow KCl + O_2$

Answer: $4KClO_3 \rightarrow 4KCl + 6O_2$. Both sides have 4 K, 4 Cl and 12 O atoms.

Question: In-text question: Balance the following equation: $CaCO_3 + HCl \rightarrow CaCl_2 + H_2O + CO_2$

Answer: $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$.

Question: In-text question: Prove that Avogadro's number is related to the

mole of a substance.

Answer: One mole of any substance contains $6.02 \times 10^{23}$ particles, and this very number is called Avogadro's number (NA), so the mole and Avogadro's number are linked.

Question: In-text question: Find the number of moles in 30 g of H3PO4.

Answer: Molar mass of H3PO4 = 3×1 + 31 + 4×16 = 98 g.Moles =

= 30 ÷ 98

= 0.31 mol.

## ائتم جي بناوت2باب :

Question: What is the total number of protons and neutrons in an atom called? (a) Number of electrons (b) Nucleon number (c) Atomic number of the element (d) Number of isotopes

Answer: Correct answer: (b) Nucleon number, also called the mass number (A).

Question: Potassium has proton number 19. What is its electron arrangement? (a) 2, 8, 9 (b) 2, 8, 8, 1 (c) 2, 8, 1 (d) 2, 8, 3

Answer: Correct answer: (b) 2, 8, 8, 1, because the 19 electrons are distributed in the K, L, M shells in this way.

Question: Potassium has nucleon number 39. How many neutrons does it have? (a) 39 (b) 19 (c) 20 (d) 29

Answer: Correct answer: (c) 20, because neutrons = mass number − proton number = 39 − 19 = 20.

Question: What is the abundance of the carbon-12 isotope? (a) 96.9% (b) 97.6% (c) 98.89% (d) 99.7%

Answer: Correct answer: (c) 98.89%. [Note: the chapter text gives the abundance of carbon-12 as 98.98%, while the question's option shows 98.89; (c) is the closest intended option.]

Question: The arrangement of electrons belongs to which particle? (a) Proton (b) Neutron (c) Electron (d) Positron

Answer: Correct answer: (c) Electron, because it is the electrons that are arranged in shells and subshells.

Question: Which particle inside the nucleus has no charge? (a) Electron (b) Proton (c) Alpha (d) Neutron

Answer: Correct answer: (d) Neutron, which carries no charge.

Question: How many subshells are there in the L shell? (a) One (b) Two (c) Three (d) Four

Answer: Correct answer: (b) Two, namely 2s and 2p.

Question: In which year was de Broglie's hypothesis about the wave nature of the electron proposed? (a) 1920 (b) 1922 (c) 1923 (d) 1925

Answer: Correct answer: (c) 1923.

Question: What material was used in Rutherford's atomic model experiment? (a) Aluminium foil (b) Zinc sulphide (c) Sodium sulphide (d) Aluminium sulphide

Answer: Correct answer: (b) Zinc sulphide (ZnS), on whose screen the alpha particles struck and produced flashes.

Question: Which rays are used to sterilise medical instruments? (a) Alpha rays (b) Beta rays (c) Gamma rays (d) X-rays

Answer: Correct answer: (c) Gamma rays, because they carry very high energy and kill germs.

Question: Define isotopes and write the structure of the isotopes of chlorine.

Answer: Isotopes are atoms of the same element with the same proton number but different nucleon numbers. Isotopes of chlorine: $^{35}Cl_{17}$ (17 protons, 18 neutrons) and $^{37}Cl_{17}$ (17 protons, 20 neutrons).

Question: An atom has 5 electrons in its M shell. Find: (a) its atomic number (b) its electronic configuration (c) the name of the element

Answer: (a) Atomic number =

= 2 + 8 + 5

= 15. (b) Electronic configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$ or 2, 8, 5. (c) Element: phosphorus (P).

Question: Prove with reasons that Rutherford's atomic model has defects.

Answer: (1) It could not explain how the positive charge stays together in the nucleus. (2) Orbiting electrons should lose energy and fall into the nucleus, yet the atom is stable. (3) It could not explain the line spectrum of the atom.

Question: State de Broglie's hypothesis about the wave-particle nature of the electron.

Answer: Every moving particle also has wave-like properties. The electron too behaves like a wave. Wavelength: $\lambda = h/mv$, where h is Planck's constant, m is mass and v is velocity.

Question: What are the defects of Bohr's atomic model?

Answer: (1) It applies only to single-electron particles. (2) It does not explain the fine structure of electron orbits. (3) It goes against Heisenberg's uncertainty principle, because the position and velocity of an electron cannot both be known at the same time.

Question: Differentiate between shell and subshell, with examples.

Answer: A shell is the main energy level of electrons in an atom (K, L, M, N). A subshell is a smaller division of a shell (s, p, d, f). Example: the L shell contains the 2s and 2p subshells.

Question: How are $^{16}O$ and $^{17}O$ atoms alike and different? Explain.

Answer: Alike: both have 8 protons and 8 electrons, both are oxygen, and their chemical properties are the same. Different: $^{16}O$ has 8 neutrons and $^{17}O$ has 9 neutrons, so their masses and physical properties differ.

Question: Write the names of the subatomic particles with mass and charge.

Answer: Proton: mass $1.67\times10^{-27}$ kg, charge +1. Neutron: mass $1.67\times10^{-27}$ kg, charge 0. Electron: mass $9.1\times10^{-31}$ kg, charge −1.

Question: Review atomic structure in light of Rutherford's gold-foil

experiment.

Answer: Rutherford fired alpha particles at a thin gold foil. Most passed straight through, some were deflected and a few bounced back. Conclusion: the atom is mostly empty, the positive charge and mass are gathered in a tiny nucleus, and electrons revolve around it.

Question: State the uses of isotopes in daily life.

Answer: (1) Carbon-14 is used to date old objects. (2) Cobalt-60 is used in cancer treatment. (3) Iodine-131 is used to examine thyroid disease. (4) Uranium-235 is used as fuel in nuclear power stations.

Question: How is Bohr's model different from Rutherford's model? Explain.

Answer: According to Rutherford, electrons can revolve in any orbit, whereas according to Bohr, electrons revolve only in fixed-energy orbits and absorb or release energy when changing orbit. Bohr is based on the quantum idea; Rutherford on the classical idea.

Question: Prove the relation of de Broglie's modern theory with Einstein's and Planck's equations.

Answer: Planck: $E = h\nu$ and Einstein: $E = mc^2$. Equating them gives $h\nu = mc^2$. Putting $\nu = c/\lambda$ gives $h/\lambda = mc$, so $\lambda = h/mc$. For a moving particle $\lambda = h/mv$, which is de Broglie's equation.

Question: How are cathode rays obtained? What are their main properties?

Answer: Cathode rays are obtained by passing electricity through a gas at low pressure and high voltage in a discharge tube. Properties: they travel in straight lines, carry negative charge, bend in electric and magnetic fields, and can pass through metal foil.

Question: Describe Schrödinger's atomic model.

Answer: According to Schrödinger, electrons are not in fixed orbits but in orbitals, where there is a probability of finding them. Atomic structure is described by the wave equation. This model is based on quantum

mechanics.

Question: Describe in detail the experiments proving the existence of electrons, protons and neutrons in the atom.

Answer: Electron: the cathode-ray experiment (Crookes and Thomson). Proton: Goldstein's canal-ray experiment. Neutron: Chadwick's experiment, in which alpha particles were fired at beryllium and chargeless particles came out.

Question: How many protons, neutrons and electrons are in: (1) $^{56}Fe_{26}$ (2) $^{17}O_8$ (3) $^{37}Cl_{17}$ (4) $^{235}U_{92}$ (5) $^{14}C_6$

Answer: (1) $^{56}Fe_{26}$: protons 26, electrons 26, neutrons 30. (2) $^{17}O_8$: protons 8, electrons 8, neutrons 9. (3) $^{37}Cl_{17}$: protons 17, electrons 17, neutrons 20. (4) $^{235}U_{92}$: protons 92, electrons 92, neutrons 143. (5) $^{14}C_6$: protons 6, electrons 6, neutrons 8.

Question: In-text question: An oxygen atom has 8 protons and 8 electrons. What is its atomic number?

Answer: Atomic number =

= number of protons

= 8.

Question: In-text question: A chlorine atom has 17 protons and 18 neutrons. Find its mass number.

Answer: Mass number =

= protons + neutrons

= 17 + 18

= 35.

Question: In-text question: How many protons, electrons and neutrons are in an atom of $^{59}Co_{27}$?

Answer: Protons = 27, electrons = 27, neutrons = 59 − 27 = 32.

Question: In-text question: Do you know any element whose atom has no neutron?

Answer: Yes, the hydrogen atom (protium $^{1}H$) has no neutron.

Question: In-text question: Which particles show the mass of the atom?

Answer: Protons and neutrons, which are in the nucleus and make up almost all the mass of the atom.

Question: In-text question: Establish Rutherford's atomic model based on classical theory and Bohr's atomic model based on quantum theory.

Answer: Rutherford's model is based on the classical idea: electrons can revolve in any orbit. Bohr's model is based on the quantum idea: electrons revolve only in fixed-energy orbits and take in or give out packets (quanta) of energy.

Question: In-text question: How can you show the relation of everyday things with chemistry?

Answer: Every everyday thing is made of atoms and molecules, and its properties depend on its chemical composition; for example food, water and medicines are all chemical substances.

Question: In-text question: How many electrons can an s subshell hold?

Answer: An s subshell can hold a maximum of 2 electrons.

Question: In-text question: How many electrons are in the L shell of an atom with atomic number 11?

Answer: Sodium (Z = 11) has the arrangement 2, 8, 1, so there are 8 electrons in the L shell.

Question: In-text question: Why do electrons fill shells in order?

Answer: Inner shells have lower energy, so electrons first fill the lower-energy shells.

Question: In-text question: An atom has its K and L shells completely full. How many electrons are there in total?

Answer: 2 in the K shell and 8 in the L shell, so 10 electrons in total.

Question: In-text question: How many neutrons are in the isotopes of hydrogen?

Answer: Protium has 0, deuterium has 1 and tritium has 2 neutrons.

Question: In-text question: Why do isotopes of an element have the same chemical but different physical properties?

Answer: Chemical properties depend on the electron arrangement, which is the same, while physical properties depend on the mass number, which differs.

Question: In-text question: How are the isotopes of carbon different from the isotopes of hydrogen?

Answer: In the isotopes of carbon, all have 6 protons and 6, 7 or 8 neutrons. In the isotopes of hydrogen, there is 1 proton and 0, 1 or 2 neutrons.

## باب3ون خاصيتون دوري جدول دوري :

Question: Mendeleev (1869) presented his periodic law that the properties of elements are a periodic function of: (a) Atomic number (b) Chemical properties (c) Physical properties (d) Atomic mass

Answer: Correct answer: (d) Atomic mass, because Mendeleev arranged the elements in increasing order of atomic mass.

Question: On what basis is the periodic table divided into s, p, d, f blocks? (a) Atomic radius (b) Electronic configuration (c) Ionisation energy (d) Electron affinity

Answer: Correct answer: (b) Electronic configuration, i.e. which subshell the last electron enters.

Question: What are the 4th and 5th periods called? (a) Short periods (b) Long periods (c) Normal periods (d) All very long periods

Answer: Correct answer: (b) Long periods, because they contain 18 elements each.

Question: What decreases from left to right in a period? (a) Ionisation energy (b) Atomic radius (c) Electronegativity (d) Electron affinity

Answer: Correct answer: (b) Atomic radius, because the nuclear pull increases while the shells stay the same.

Question: What are the Group VIIA elements called? (a) Lanthanides (b) Actinides (c) Halogens (d) Inert or noble gases

Answer: Correct answer: (c) Halogens, such as fluorine, chlorine, bromine and iodine.

Question: According to Moseley, the chemical properties of elements are a periodic function of: (a) Atomic size (b) Atomic mass (c) Atomic radius (d) Atomic number

Answer: Correct answer: (d) Atomic number, which is the basis of the modern periodic law.

Question: How does the shielding effect change across a period? (a) Increases (b) Decreases (c) Increases then stays (d) Stays the same

Answer: Correct answer: (b) Decreases, because the nuclear charge increases while the inner electrons stay the same.

Question: What is the attraction for a shared pair of electrons called? (a) Electron affinity (b) Electronegativity (c) Ionisation energy (d) Shielding effect

Answer: Correct answer: (b) Electronegativity.

Question: Why does electron affinity decrease down a group? (a) Atomic size increases (b) Atomic size decreases (c) Atomic size decreases then increases (d) Atomic size increases then stays

Answer: Correct answer: (a) Atomic size increases, so the nuclear pull on the outer electron decreases.

Question: What type are the transition and inner-transition elements? (a) Gases (b) Metals (c) Non-metals (d) Metalloids

Answer: Correct answer: (b) Metals, such as iron, copper and zinc.

Question: What is the difference between a period and a group?

Answer: A period is a horizontal row of the periodic table in which elements are in increasing atomic number. A group is a vertical column whose elements have equal electrons in the outer shell and similar properties.

Question: Explain the trend of electronegativity in a group and in a period, with examples.

Answer: Down a group, electronegativity decreases (F = 4.0, Cl = 3.0, Br = 2.8). Across a period from left to right, electronegativity increases (Li = 1.0, C = 2.5, F = 4.0).

Question: Explain the similarity of the physical and chemical properties of elements of the same family.

Answer: Elements of the same group have equal electrons in the outer shell, so their chemical properties are similar and their physical properties change in order. Example: the alkali metals (Li, Na, K) are all soft and reactive.

Question: Prove that the periodic properties of an element depend on the number of protons in its atom.

Answer: The number of protons = the atomic number, which fixes the element's place in the table. The atomic number fixes the electronic configuration, and the electronic configuration fixes the periodic properties.

Question: Which halogen elements are found in gas, liquid and solid states? Identify them.

Answer: Fluorine and chlorine are gases, bromine is a liquid and iodine is a solid.

Question: Which alkaline-earth metal forms a coating over another metal?

Answer: Magnesium (Mg) forms a protective oxide layer on its surface. [Note: the printed wording of the question is not fully clear; this answer is a best effort based on the chapter content.]

Question: What trends do ionisation energy, electron affinity and electronegativity show in a group and in a period?

Answer: Down a group, all three decrease. Across a period from left to right, all three increase, because the nuclear pull increases and the atomic size decreases.

Question: Explain the periodic table in detail.

Answer: The periodic table is an ordered list of the elements with 7 periods (horizontal rows) and 18 groups (vertical columns). The elements are

arranged in increasing atomic number. The table is divided into s, p, d, f blocks, and the groups are named by family, such as alkali metals, alkaline-earth metals, halogens and noble gases.

Question: Classify the periodic table into s, p, d, f blocks by degree.

Answer: s block: Groups IA and IIA (last electron in s). p block: Groups IIIA to VIIIA (last electron in p). d block: transition elements (last electron in d). f block: lanthanides and actinides (last electron in f).

Question: Show the electronic configuration of Na, Ca, F, Si.

Answer: Na (Z=11): $1s^2\ 2s^2\ 2p^6\ 3s^1$ or 2, 8, 1. Ca (Z=20): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$ or 2, 8, 8, 2. F (Z=9): $1s^2\ 2s^2\ 2p^5$ or 2, 7. Si (Z=14): $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$ or 2, 8, 4.

Question: Determine the position of families in the periodic table.

Answer: Families are the vertical groups: IA alkali metals, IIA alkaline-earth metals, IIIA to VIA representative elements, VIIA halogens, VIIIA noble gases, and IB to VIIIB transition metals in the middle.

Question: On what basis did Mendeleev lay the foundation of the modern periodic law? Discuss.

Answer: Mendeleev arranged the elements in increasing atomic mass and grouped elements of similar properties together. He left gaps for undiscovered elements and predicted their properties, which later proved correct. On this basis the modern law is founded on atomic number.

Question: Explain the shielding effect and how it affects the atomic radius.

Answer: The shielding effect is that the inner electrons hide the nuclear pull from the outer electrons, so the outer electrons feel less pull. The greater the shielding, the weaker the nuclear effect and the larger the atomic radius.

Question: In-text question: Answer the following questions with the help of the periodic table: make a list of the melting points of solid, liquid and gas

elements.

Answer: From the periodic table it can be seen that most metals are solids with high melting points, bromine is a liquid, and non-metals such as hydrogen and oxygen are gases.

Question: In-text question: Identify the names of the elements from the periodic table and write their names.

Answer: In the periodic table, each cell carries the element's symbol and name; for example H is hydrogen, O is oxygen, Fe is iron and Au is gold.

Question: In-text question: Show the atomic number of an element from the periodic table.

Answer: The atomic number is written at the top of each element's cell; for example carbon has atomic number 6 and sodium 11.

Question: In-text question: Identify the alkali, alkaline-earth and transition elements from the periodic table.

Answer: Alkali metals: Group IA (Li, Na, K). Alkaline-earth metals: Group IIA (Mg, Ca, Ba). Transition metals: the middle groups IB to VIIIB (Fe, Cu, Zn).

Question: In-text question: Identify the names of the lanthanide and actinide series elements from the periodic table.

Answer: Lanthanides: from lanthanum (La) to lutetium (Lu). Actinides: from actinium (Ac) to lawrencium (Lr).

Question: In-text question: What is the trend of atomic radius in a group?

Answer: Down a group, the atomic radius increases, because new shells keep being added.

Question: In-text question: What is the shielding effect on an atom?

Answer: The shielding effect is that the inner electrons block the nuclear pull from the outer electrons, so the outer electrons feel less pull.

Question: In-text question: Which element has the highest ionisation energy and why?

Answer: Helium has the highest ionisation energy, because its atom is the smallest and its electrons are held very tightly by the nucleus.

## کیمیائي بانڈنگ4باب :

Question: Which of the following is an example of an ionic compound? (a) H2 (b) CH4 (c) N2 (d) NaCl

Answer: Correct answer: (d) NaCl, because it forms by the transfer of electrons.

Question: What is the mutual interaction between a hydrogen atom bonded to a highly electronegative atom and another electronegative atom called? (a) Covalent bond (b) Ionic bond (c) Hydrogen bond (d) Metallic bond

Answer: Correct answer: (c) Hydrogen bond.

Question: By sharing one electron, a fluorine atom gets the configuration of which element? (a) Xe (b) Ar (c) Kr (d) Ne

Answer: Correct answer: (d) Ne, because $F^-$ then has 10 electrons, like neon.

Question: How many electrons does a Group IIIA atom give up? (a) 1 (b) 2 (c) 3 (d) 4

Answer: Correct answer: (c) 3, because its outer shell has 3 electrons.

Question: Which atom gives 2 electrons from its outer shell? (a) Oxygen (b) Potassium (c) Magnesium (d) Carbon

Answer: Correct answer: (c) Magnesium, whose arrangement is 2, 8, 2 and which forms $Mg^{2+}$.

Question: In the NaCl crystal structure, each $Na^+$ is surrounded by how many $Cl^-$ ions? (a) 6 $Cl^-$ (b) 6 $Na^+$ (c) 8 $Cl^-$ (d) 12 $Cl^-$

Answer: Correct answer: (a) 6 $Cl^-$ ions.

Question: In what state are ionic compounds mostly found at room temperature? (a) Gas (b) Crystalline solid state (c) Liquid (d) Plasma

Answer: Correct answer: (b) Crystalline solid state.

Question: What is an atom's tendency to gain electrons in its outer shell called? (a) Octet rule (b) Duplet rule (c) Triplet rule (d) None

Answer: Correct answer: (a) Octet rule, i.e. completing 8 electrons in the outer shell.

Question: When one atom donates an electron pair and the other accepts it, which bond is formed? (a) Covalent bond (b) Ionic bond (c) Coordinate covalent bond (d) Hydrogen bond

Answer: Correct answer: (c) Coordinate covalent bond.

Question: Why are noble gases stable? Because their valence shell has how many electrons? (a) 4 (b) 6 (c) 8 (d) 10

Answer: Correct answer: (c) 8 electrons (2 in helium), so they need neither gain nor lose electrons.

Question: What is a bond called in which three electron pairs are shared? (a) Double covalent bond (b) Single covalent bond (c) Triple covalent bond (d) None

Answer: Correct answer: (c) Triple covalent bond, such as in N2.

Question: How does a non-metal atom get a negative charge? (a) By losing electrons (b) By gaining electrons (c) By losing protons (d) By gaining protons

Answer: Correct answer: (b) By gaining electrons, e.g. $Cl + e^- \rightarrow Cl^-$.

Question: When two atoms share an electron pair equally, what is the bond called? (a) Non-polar covalent bond (b) Polar covalent bond (c) Double covalent bond (d) Coordinate covalent bond

Answer: Correct answer: (a) Non-polar covalent bond, such as in H2 and Cl2. [Note: this is the best reading of the printed wording of the question.]

Question: Adhesives are used where what is needed? (a) Electrical insulation/barrier (b) Water barrier (c) Attraction of particles (d) Colour

Answer: Correct answer: (a) Electrical insulation/barrier, because the chapter's STS section mentions epoxy adhesives used in electronics. [Note: the printed wording of the question is not fully clear.]

Question: How many electrons are in the valence shell of oxygen (Group VIA)? (a) 3 (b) 4 (c) 5 (d) 6

Answer: Correct answer: (d) 6 electrons.

Question: Which electron pair does an atom contribute in bonding? (a) Electron pair (b) Lone pair (c) Bond pair (d) Shared pair

Answer: Correct answer: (d) Shared pair, which is also called the bond pair. [Note: (c) and (d) are near-synonyms; (d) is the printed answer.]

Question: What are ionic crystals like? (a) High melting point (b) Medium melting point (c) Low melting point (d) None

Answer: Correct answer: (a) High melting point, because the attraction between ions is very strong.

Question: How is the strength of a covalent bond compared with an ionic bond? (a) Weak (b) Strong (c) Equal (d) None

Answer: Correct answer: (a) Weak, i.e. the ionic bond is stronger.

Question: Which bond forms by sharing electrons? (a) Ionic bond (b) Coordinate covalent bond (c) Covalent bond (d) Metallic bond

Answer: Correct answer: (c) Covalent bond.

Question: Which diagram shows a bond between atoms of equal electronegativity?

Answer: The diagram with no + or − marks on A–B, i.e. the non-polar bond. [Note: the four printed options look almost identical in the scan.]

Question: Show with dot-and-cross (X / •) diagrams how different types of chemical bonds form: (a) hydrogen (H2, covalent) (b) potassium fluoride (KF, ionic)

Answer: (a) H2: H• + •H → H:H; both H atoms share electrons. (b) KF: K (2,

8, 8, 1) gives its one electron to F (2, 7); $K^+$ and $F^-$ form and join by attraction.

Question: What is meant by the Octet Rule and the Duplet Rule?

Answer: Octet rule: an atom wants to complete 8 electrons in its outer shell. Duplet rule: small atoms such as hydrogen and helium want to complete 2 electrons in their shell.

Question: Show an atom that has 2 electrons in its valence shell.

Answer: Magnesium (Mg): electron arrangement 2, 8, 2, i.e. 2 electrons in the valence shell.

Question: How does oxygen get a negative (anion) charge?

Answer: Oxygen (2, 6) gains two electrons to form $O^{2-}$ and completes its octet: $O + 2e^- \rightarrow O^{2-}$.

Question: What is the difference between a lone pair and a bond pair?

Answer: A lone pair does not take part in bonding and stays on one atom. A bond pair is shared between two atoms and forms a bond.

Question: Why is the melting point of a covalent compound (like ice) low? Explain.

Answer: Between covalent molecules there are weak intermolecular forces, which break with little heat, so the melting point is low. [Note: the printed wording of the question is not fully clear.]

Question: How is the nature of a chemical bond known from electronegativity-difference values?

Answer: Difference 0 to 0.4: non-polar covalent. 0.4 to 1.7: polar covalent. More than 1.7: ionic bond.

Question: Explain why a magnesium atom readily gives two electrons.

Answer: Magnesium's arrangement is 2, 8, 2. By giving two electrons it becomes $Mg^{2+}$ and gains a stable octet.

Question: Why can metallic elements' atoms form ionic bonds but not

covalent bonds?

Answer: Metals like to give electrons, so they form ions and make ionic bonds. A covalent bond requires sharing electrons, which is not the nature of metals.

Question: What is an ion? How is it different from an atom?

Answer: An ion is a charged atom or group of atoms. An atom is neutral, whereas an ion forms by gaining or losing electrons and is positive (cation) or negative (anion).

Question: Explain the dipole force.

Answer: The dipole-dipole force is the attraction between polar molecules: the positive end of one molecule attracts the negative end of another.

Question: Write the uses of adhesives.

Answer: Adhesives are used to join broken things, to stick wood, paper, cloth and plastic, and to join parts in electronics.

Question: Why is the intermolecular force weaker than the intramolecular force?

Answer: Intramolecular forces are strong chemical bonds between atoms, whereas intermolecular forces are weak attractions between molecules, so they are weaker.

Question: What are the properties of metallic bonds?

Answer: In a metallic bond, metal atoms are joined in a sea of electrons. That is why metals are good conductors of heat and electricity, shiny, malleable and ductile.

Question: Covalent bonds are strong, yet most covalent molecules have low melting points. Why?

Answer: The bonds inside a molecule are strong, but the forces between molecules are weak. On melting, the weak forces between molecules break, not the inner bonds, so the melting point is low.

Question: Show the properties of ionic compounds.

Answer: (1) Crystalline solids. (2) High melting and boiling points. (3) Conductors of electricity in the molten or dissolved state. (4) Hard and brittle.

Question: What are ionic compounds?

Answer: Ionic compounds are compounds formed by the attraction between positive and negative ions, such as NaCl and MgO.

Question: How does hydrogen bonding affect the physical properties of compounds? Explain.

Answer: Hydrogen bonding holds molecules close together, so melting and boiling points rise, solubility in water increases, and special properties such as those of water appear.

Question: Complete the chart: the given row for Z = 11 (11, 11, 2, 8, 1, 1) and for Z = 12 to 16 write the atomic number, protons, electrons, electronic arrangement and valence electrons.

Answer: 12 Mg: 12, 12, 2, 8, 2, 2. 13 Al: 13, 13, 2, 8, 3, 3. 14 Si: 14, 14, 2, 8, 4, 4. 15 P: 15, 15, 2, 8, 5, 5. 16 S: 16, 16, 2, 8, 6, 6.

Question: Describe the ionic bond, with the formation of NaCl.

Answer: An ionic bond forms by the transfer of electrons. Na (2, 8, 1) gives its one electron to become $Na^+$, Cl (2, 8, 7) takes an electron to become $Cl^-$. The attraction of opposite charges forms NaCl.

Question: How do elements gain stability?

Answer: Elements gain stability by completing their outer shell: by giving, taking or sharing electrons. Metals give electrons; non-metals take or share electrons, so that the octet or duplet is completed.

Question: Explain how ionic and covalent bonds form between atoms.

Answer: Ionic bond: one atom gives electrons, the other takes them, ions form and join by attraction. Covalent bond: both atoms share electron

pairs, so that both complete their octets.

Question: Show the electron arrangement of these molecules with dot-cross diagrams: (a) $H_2O$ (b) $N_2$ (c) $CH_4$ (d) $C_2H_2$ (e) $Cl_2$ (f) $H_2$

Answer: (a) $H_2O$: O in the centre, single bonds to two H, two lone pairs on O. (b) $N_2$: N≡N triple bond, one lone pair on each N. (c) $CH_4$: C in the centre, single bonds to 4 H. (d) $C_2H_2$: H–C≡C–H. (e) $Cl_2$: Cl–Cl single bond. (f) $H_2$: H–H single bond.

Question: Define metallic bond and explain its structure/nature.

Answer: A metallic bond is the bond between metal atoms in which the valence electrons are shared by all atoms and form a sea of electrons. Positive metal ions are held in this sea, which makes metals conductors and malleable.

Question: What is a coordinate covalent bond? Explain with examples.

Answer: A coordinate covalent bond is a bond in which the shared electron pair is given by only one atom. Examples: in $NH_4^+$, N gives its lone pair to $H^+$. In $H_3O^+$, O gives its lone pair to $H^+$.

Question: What is meant by the ionic character of a covalent bond?

Answer: When the electronegativity of two atoms differs, the shared electron pair is pulled towards the more electronegative atom, so the bond gains partial ionic character. The greater the difference, the greater the ionic character.

Question: Differentiate between polar and non-polar compounds in properties.

Answer: Polar compounds: partially charged, soluble in water, higher melting and boiling points. Non-polar compounds: uncharged, insoluble in water, lower melting and boiling points.

Question: Explain the importance of adhesives and epoxy adhesives in society.

Answer: Adhesives are used to join broken things, stick wood and paper, and for daily repairs. Epoxy adhesives are very strong and are used in electronics, aircraft and buildings.

Question: In-text question: In which group is magnesium? How many electrons are in the outer shell of a magnesium atom?

Answer: Magnesium is in Group IIA and has 2 electrons in its outer shell (2, 8, 2).

Question: In-text question: What is the charge on the magnesium ion and what is its symbol?

Answer: The magnesium ion carries a +2 charge and its symbol is $Mg^{2+}$.

Question: In-text question: Why is the fluoride ion considered like neon?

Answer: The fluoride ion ($F^-$) has 10 electrons, the same as neon (Ne), so its arrangement is neon-like.

Question: In-text question: Is the bond of a negatively charged ion called dipolar?

Answer: When opposite partial charges appear on the two sides of a bond, that bond becomes dipolar (polar). [Note: the printed wording of the question is not fully clear.]

Question: In-text question: Why does a helium atom not gain electrons?

Answer: The helium shell already has 2 electrons and its duplet is complete, so it is stable and does not gain electrons.

Question: In-text question: Show with cross (X) and dot (•) the formation of anions in non-metal atoms: (a) sulphur (atomic number 16) (b) oxygen (atomic number 8)

Answer: (a) $S + 2e^- \rightarrow S^{2-}$, arrangement 2, 8, 8. (b) $O + 2e^- \rightarrow O^{2-}$, arrangement 2, 8.

Question: In-text question: How many electrons are in the valence shell and why are they important?

Answer: A valence shell holds at most 8 electrons. They are important because they take part in bonding and chemical reactions.

Question: In-text question: Why do noble gases not react with other elements?

Answer: The outer shells of noble gases are complete (octet or duplet), so they are stable and do not react.

Question: In-text question: Show with cross-dot the formation of cations in metal atoms: (a) K (atomic number 19) (b) aluminium (atomic number 13)

Answer: (a) K (2, 8, 8, 1) → $K^+$ (2, 8, 8) + $e^-$. (b) Al (2, 8, 3) → $Al^{3+}$ (2, 8) + $3e^-$.

Question: In-text question: Is a coordinate covalent bond always a polar bond?

Answer: Yes, because the shared electron pair is given by one atom only, so partial charges separate in the bond and the bond has polar character.

Question: In-text question: Show the dot-cross structure of CCl4.

Answer: Carbon is in the centre and forms single bonds to 4 chlorines. Each chlorine has 3 lone pairs and carbon's octet is complete.

Question: In-text question: Why can a hydrogen atom not form more than one covalent bond?

Answer: Hydrogen has only 1 electron and needs only one more electron to complete its duplet, so it forms only one bond.

Question: In-text question: Show with dot-cross the formation of anions from non-metal atoms: (a) N (b) Br (c) P

Answer: (a) N + $3e^-$ → $N^{3-}$. (b) Br + $e^-$ → $Br^-$. (c) P + $3e^-$ → $P^{3-}$.

Question: In-text question: In which molecule does polarity occur?

Answer: In a molecule in which the atoms differ in electronegativity, such as HCl, H2O and NH3, polarity occurs.

Question: In-text question: What is meant by the intermolecular force

between molecules?

Answer: The intermolecular force is the weak attraction that holds neighbouring molecules together.

Question: In-text question: Why are metals good conductors of heat and electricity?

Answer: Metals have free/shared electrons which move easily and carry heat and electricity from one place to another.

## مادي جون طبعي حالتون5باب :

Question: Which gas diffuses fastest? (a) Hydrogen (b) Chlorine (c) Fluorine (d) Helium

Answer: Correct answer: (a) Hydrogen, because it is the lightest (molecular mass = 2).

Question: The vapour pressure of water increases with: (a) Pressure (b) Temperature (c) Intermolecular force (d) Molecular polarity

Answer: Correct answer: (b) Temperature, because as temperature rises, more molecules become vapour.

Question: On what does the boiling point depend? (a) Nature of the liquid (b) Pressure (c) Temperature (d) Volume

Answer: Correct answer: (b) Pressure, because the boiling point is directly proportional to atmospheric pressure. [Note: the chapter lists two factors affecting boiling point: atmospheric pressure and the nature of the liquid; the question intends pressure.]

Question: One atmosphere pressure equals how many pascal? (a) 10325 (b) 106075 (c) 10523 (d) 101325 pascal

Answer: Correct answer: (d) 101325 pascal.

Question: Which of the following does NOT affect the rate of evaporation? (a) Intermolecular forces (b) External pressure (c) Initial temperature of the liquid (d) Nature of the liquid

Answer: Correct answer: (b) External pressure, because evaporation happens at the surface and pressure in open space does not affect it much.

Question: In which state are the slowest-moving particles? (a) Solid (b) Gas (c) Plasma (d) Bose-Einstein Condensate

Answer: Correct answer: (a) Solid, because in a solid the particles only vibrate in place.

Question: Which solid has a definite melting point? (a) Plastic (b) Rubber (c) Glass (d) Diamond

Answer: Correct answer: (d) Diamond, because it is a crystalline solid.

Question: Which is the lightest matter among the following? (a) Solid (b) Liquid (c) Gas (d) Plasma

Answer: Correct answer: (c) Gas, because gas particles are very far apart and its density is low.

Question: In evaporation, why do molecules leave the liquid surface? (a) They have less energy (b) They have more energy (c) They have equal energy (d) None

Answer: Correct answer: (b) They have more energy, so they escape the intermolecular attraction and leave the surface.

Question: When does a gas expand? (a) When pressure increases (b) When temperature increases (c) When volume increases (d) None

Answer: Correct answer: (b) When temperature increases (at constant pressure), according to Charles's law.

Question: Define allotropy, with examples.

Answer: Allotropy is the property in which the same element occurs in different forms. Example: the allotropes of carbon are diamond, graphite and fullerenes.

Question: What are effusion and diffusion? Give examples.

Answer: Diffusion: the spontaneous spreading of gas or liquid molecules, such as perfume spreading. Effusion: gas escaping through a tiny hole, such as gas leaking from a balloon.

Question: Describe: freezing point, melting point, boiling point.

Answer: Freezing point: the temperature at which a liquid becomes a solid.

Melting point: the temperature at which a solid becomes a liquid. Boiling point: the temperature at which the vapour pressure of a liquid equals atmospheric pressure and the liquid starts to boil.

Question: What is evaporation? How do temperature and pressure affect it?

Answer: Evaporation is the process in which liquid molecules escape from the surface as vapour at any temperature. Evaporation increases with temperature. Pressure has little direct effect on evaporation, but in a closed system the vapour pressure rises.

Question: Explain plasma, with daily-life examples.

Answer: Plasma is the fourth state of matter in which gas atoms split into ions and free electrons. Examples: lightning, neon lights, TV screens and plasma lamps.

Question: Prove with reasons that Bose-Einstein Condensate is super-cold and super-unexcited matter.

Answer: Bose-Einstein Condensate forms near absolute zero (0 K, −273°C). At such cold, the motion of atoms almost stops, so it is super-cold and super-unexcited matter.

Question: How does kinetic molecular theory show the difference in the states of matter?

Answer: According to this theory, in a solid the particles are close and only vibrate, in a liquid the particles can flow over one another, and in a gas the particles are far apart and move freely. As temperature rises, the motion of the particles increases.

Question: What is evaporation? What factors affect evaporation? Why does evaporation cause cooling?

Answer: Evaporation is the escape of molecules from a liquid surface as vapour. Factors affecting it: temperature, surface area, air movement,

intermolecular forces and the nature of the liquid. Evaporation causes cooling because the higher-energy molecules leave and the average energy of the remaining molecules falls.

Question: Explain Boyle's law, with examples.

Answer: Boyle's law: at constant temperature, the volume of a gas is inversely proportional to its pressure, i.e. PV = K. Example: pressing air in a syringe decreases its volume and increases its pressure.

Question: What is the difference between crystalline and amorphous solids?

Answer: Crystalline solids: particles in order, definite melting point, e.g. diamond, salt. Amorphous solids: particles in disorder, no definite melting point, e.g. glass, plastic.

Question: State Charles's law, in definition and detail.

Answer: Charles's law: at constant pressure, the volume of a gas is directly proportional to its absolute temperature, i.e. V/T = K. Example: warming a balloon makes it expand.

Question: Explain diffusion in liquids; what effects/factors apply?

Answer: Diffusion in liquids is the spreading of the molecules of one liquid through another, such as ink spreading in water. It is affected by temperature, particle size and the density of the liquid.

Question: How do different melting points affect solids? Explain.

Answer: Solids with higher melting points have stronger bonds and melt slowly on heating, such as diamond. Those with lower melting points melt quickly, such as ice and wax. [Note: this is the best reading of the printed wording of the question.]

Question: Explain vapour pressure and prove it occurs only in a closed system.

Answer: Vapour pressure is the pressure of a liquid's vapour above the

liquid. In a closed system the vapour cannot escape, so equilibrium is set up between evaporation and condensation and the pressure becomes constant. In an open system the vapour flies away, so no such equilibrium is set up.

Question: Convert: (a) 100°C to K (b) 150°C to K (c) 750 K to °C (d) 170 K to °C

Answer: (a) 100 + 273 = 373 K. (b) 150 + 273 = 423 K. (c) 750 − 273 = 477°C. (d) 170 − 273 = −103°C.

Question: At constant pressure, a gas expands from 90.5 cubic cm to 120 cubic cm; the initial temperature is 33°C. What will the final temperature be?

Answer: T1 =

= 33 + 273

= 306 K. By Charles's law V1/T1 = V2/T2, so T2 = 120 × 306 ÷ 90.5 = 405.6 K = 132.6°C. [Note: the printed unit is cm², read as cm³.]

Question: A gas of 78 ml at constant pressure is heated from 35°C to 80°C. What will the final volume be?

Answer: T1 = 35 + 273 = 308 K, T2 = 80 + 273 = 353 K.V2 =

= V1 × T2 ÷ T1

= 78 × 353 ÷ 308

= 89.4 ml.

Question: At STP (0°C, 1 atm) a gas occupies 40.0 dm³; at constant temperature the pressure becomes 3 atm. What will the new volume be?

Answer: By Boyle's law P1V1 = P2V2, so V2 = 1 × 40 ÷ 3 = 13.33 dm³.

Question: A gas of 800 cm³ is at 750 mm Hg pressure; it is compressed to 250 cm³. What will the new pressure be?

Answer: P1V1 = P2V2, so P2 = 800 × 750 ÷ 250 = 2400 mm Hg.

Question: A gas sample at 8 atm has a volume of 15 litres; the pressure is brought to 6 atm. What will the new volume be?

Answer: P1V1 = P2V2, so V2 = 8 × 15 ÷ 6 = 20 litres.

Question: In-text question: Explain the diffusion of gases.

Answer: Diffusion is the spontaneous spreading of gas molecules due to random motion, such as perfume spreading in a room. Lighter gases spread faster.

Question: In-text question: What happens to the density of a gas when it is heated?

Answer: On heating, the gas expands and its volume increases, so its density decreases.

Question: In-text question: Explain effusion and diffusion and what they depend on.

Answer: Effusion: gas escaping through a tiny hole. Diffusion: the spreading of molecules. Both depend on molar mass (Graham's law), temperature and pressure.

Question: In-text question: What is the variable in Boyle's law?

Answer: In Boyle's law, pressure (P) and volume (V) are the variables, while temperature stays constant.

Question: In-text question: What effect does temperature have on the volume of a gas?

Answer: As temperature rises, the volume of a gas increases (Charles's law).

Question: In-text question: What is Absolute Zero?

Answer: Absolute zero is −273°C (0 K), the temperature at which the volume of a gas would become zero and molecular motion almost stops.

Question: In-text question: Can you show the relation between the volume of a gas and its temperature?

Answer: Yes, by Charles's law V ∝ T, i.e. V1/T1 = V2/T2. On a graph this relation is a straight line.

Question: In-text question: Why does a rise in temperature increase evaporation?

Answer: As temperature rises, more molecules gain enough energy to escape from the surface as vapour.

Question: In-text question: Why is the rate of diffusion in liquids lower than in gases? Explain with examples.

Answer: In a liquid the molecules are close and held by strong attraction, so they move slowly. Example: ink spreads slowly in water, while perfume spreads quickly in air.

Question: In-text question: How does the state of equilibrium in vapour pressure occur in a closed system?

Answer: In a closed system the vapour cannot escape, so the rate of evaporation becomes equal to the rate of condensation and the pressure becomes constant; this is called dynamic equilibrium.

Question: In-text question: How does atmospheric pressure affect the boiling point of a substance?

Answer: When atmospheric pressure rises, the boiling point rises; when it falls, the boiling point falls; for example, on mountains water boils below 100°C.

## گاربا6 : (Solutions)

Question: An alloy is a homogeneous mixture of: (a) Two solids (b) Two liquids (c) Two gases (d) Two metalloids

Answer: Correct answer: (a) Two solids, i.e. a homogeneous mixture of two or more metals.

Question: Heating a saturated KCl solution makes it: (a) Unsaturated (b) More saturated (c) Saturated again (d) Crystallised

Answer: Correct answer: (a) Unsaturated, because on heating more solute can dissolve.

Question: When a solute dissolves in a solvent, the mixture is called: (a) Solution (b) Mixture (c) Colloid (d) Suspension

Answer: Correct answer: (a) Solution.

Question: Solubility is shown as grams of solute per how many grams of solvent? (a) 1000 (b) 100 (c) 10000 (d) 10

Answer: Correct answer: (b) 100 grams of solvent.

Question: What is an example of a heterogeneous mixture? (a) Water and sugar (b) Water and sand (c) Water and salt (d) Water and alcohol

Answer: Correct answer: (b) Water and sand, because sand does not dissolve in water.

Question: How many grams are 2 moles of NaCl? (a) 58.5 g (b) 117 g (c) 29.25 g (d) 117 g

Answer: Correct answer: (d) 117 g, because 1 mole of NaCl = 58.5 g, so 2 moles = 2 × 58.5 = 117 g. [Note: (b) and (d) show the same value; (d) is the printed correct answer.]

Question: What is the molarity when 40 g NaCl is dissolved to make 500 $cm^3$ of solution? (a) 1.16M (b) 1.20M (c) 1.19M (d) 1.37M

Answer: Correct answer: (d) 1.37M, because moles = 40 ÷ 58.5 = 0.684, volume = 0.5 $dm^3$, so M = 0.684 ÷ 0.5 = 1.37.

Question: A 10% (m/m) sugar solution means 10 g sugar in how many grams of water? (a) 90 g water (b) 10 g water (c) 80 g water (d) 100 g water

Answer: Correct answer: (a) 90 g water, because the total solution is 100 g (10 g sugar + 90 g water).

Question: Which is an example of a true solution? (a) Starch solution (b) Soap solution (c) Ink in water (d) Toothpaste

Answer: Correct answer: (c) Ink in water. [Note: by the chapter, all four options are colloid/suspension types; none is a true solution in the book's sense. (c) is the closest classroom choice.]

Question: Which solution has the least solute? (a) 0.5M (b) 0.75M (c) 1.5M (d) 0.25M

Answer: Correct answer: (d) 0.25M, because it is the lowest molarity.

Question: When a saturated solution is cooled, it becomes: (a) Unsaturated (b) More saturated (c) Saturated (d) Supersaturated

Answer: Correct answer: (d) Supersaturated, because cooling lowers the solubility and extra solute stays in the solution.

Question: Which pair shows a solid dissolved in a liquid? (a) Gas-liquid (b) Solid-liquid (c) Liquid-gas (d) Gas-gas

Answer: Correct answer: (b) Solid-liquid, such as salt in water. [Note: the printed wording of the question is not fully clear.]

Question: In which solution are both solute and solvent solids? (a) Gas in gas (b) Solid in liquid (c) Solid in solid (d) Liquid in liquid

Answer: Correct answer: (c) Solid in solid, such as an alloy.

Question: What is the particle size in a true solution? (a) More than $10^2$ nm (b) More than $10^5$ nm (c) Less than $10^3$ nm (d) More than $10^4$ nm

Answer: Correct answer: (c) Less than $10^3$ nm; true-solution particles are smaller than 1 nm, i.e. less than $10^3$ nm. [Note: the printed term of the question is not fully clear.]

Question: Give one example for each solution type: solid in liquid

Answer: Salt in water (salt solution).

Question: Give one example for each solution type: gas in gas

Answer: Air (oxygen and other gases in nitrogen).

Question: Give one example for each solution type: solid in solid

Answer: Brass (zinc in copper).

Question: Give one example for each solution type: liquid in solid

Answer: Amalgam (silver or another metal in mercury).

Question: Give one example for each solution type: liquid in gas

Answer: Fog/mist (water droplets in air).

Question: Give one example for each solution type: liquid in liquid

Answer: Alcohol in water (alcohol solution).

Question: Explain the mutual interaction of solute and solvent in preparing a sodium chloride solution.

Answer: The polar water molecules attract the $Na^+$ and $Cl^-$ ions. The oxygen ends of water surround $Na^+$ and the hydrogen ends surround $Cl^-$, so the salt crystals break apart and dissolve. This is called hydration.

Question: What is the difference between a dilute and a concentrated solution?

Answer: A dilute solution has little solute, whereas a concentrated solution has much solute.

Question: State the properties of a solution and its components.

Answer: Properties of a solution: homogeneous mixture, very small particles, particles do not settle, light passes through. Components: solute (smaller amount) and solvent (larger amount).

Question: What is meant by "mass/volume percent (m/v)%"?

Answer: (m/v)% means grams of solute in 100 $cm^3$ of solution. Example: 5% (m/v) means 5 g solute in 100 $cm^3$ of solution.

Question: Describe a molar solution, with an example.

Answer: A molar solution is one in which 1 mole of solute is dissolved in 1 $dm^3$ of solution, called 1 M. Example: dissolving 40 g NaOH in 1 $dm^3$ of solution makes a 1 M NaOH solution.

Question: How does a colloid show the Tyndall effect?

Answer: The particles of a colloid scatter light, so the beam of light is visible in the colloid. This is called the Tyndall effect, like a car's light being visible in fog.

Question: Define the terms: (i) Dilution (ii) Concentration (iii) Solubility (iv) Molarity

Answer: (i) Dilution: making a solution more dilute, i.e. adding solvent. (ii) Concentration: the amount of solute in a given amount of solution. (iii) Solubility: the maximum amount of solute that dissolves in 100 g of solvent at a given temperature. (iv) Molarity: the moles of solute in 1 $dm^3$ of solution.

Question: Why do polar solutes dissolve only in polar solvents?

Answer: By the rule "like dissolves like", the molecules of a polar solute attract the molecules of a polar solvent, so they dissolve.

Question: Why can a polar solute not dissolve in a non-polar solvent?

Answer: The molecules of a polar solute attract one another strongly, but a non-polar solvent cannot separate them, so the solute does not dissolve.

Question: How is a solution useful for a chemist?

Answer: In solutions, chemical reactions happen quickly, solutions can be measured, standard solutions can be made, and experiments such as titration can be performed.

Question: Describe the process of "salt dissolving in water".

Answer: When salt is added to water, the water molecules surround the $Na^+$ and $Cl^-$ ions and separate them from the crystal, and the salt dissolves in water.

Question: Air, oxygen, carbon dioxide, nitrogen and hydrogen gases are one ____ solution. Which gas is the solvent in it, and why?

Answer: Air is a gas-in-gas solution. Nitrogen is the solvent, because it is present in the largest amount (about 78%).

Question: Why is petrol not soluble in water?

Answer: Petrol is non-polar and water is polar, so by the rule "like dissolves like", petrol does not dissolve in water.

Question: How is a concentrated solution made from a dilute solution? Explain.

Answer: A concentrated solution can be made from a dilute solution by evaporating solvent or by adding more solute.

Question: State the term solubility; how is the limit of solute and solvent concentration fixed?

Answer: Solubility is the maximum amount of solute that dissolves in 100 g of solvent at a given temperature. When no more solute can dissolve in the solution, the solution is saturated and that is the limit of concentration.

Question: How does salt solubility change with increase in temperature?

Answer: The solubility of most solid solutes such as salt increases with temperature, because a hot solvent can dissolve more solute.

Question: Describe the attraction of water molecules for $Na^+$ and $Cl^-$ ions.

Answer: The oxygen end of water is partially negative and the hydrogen end is partially positive. The oxygen ends attract $Na^+$ and the hydrogen ends attract $Cl^-$; this is called hydration.

Question: Explain the phrase "like dissolves like" with reference to solubility.

Answer: A polar solute dissolves in a polar solvent and a non-polar solute dissolves in a non-polar solvent, because molecules with the same kind of intermolecular forces attract one another. Example: salt in water, grease in benzene.

Question: What is the difference between a solution, a colloid and a suspension?

Answer: Solution: particles smaller than 1 nm, homogeneous, do not settle. Colloid: particles 1 to 1000 nm, scatter light, do not settle. Suspension: particles larger than 1000 nm, heterogeneous, settle down.

Question: A solution is made by dissolving 1.25 g HCl in water to 30 $cm^3$. What is the molarity?

Answer: Molar mass of HCl = 36.5 g.Moles =

= 1.25 ÷ 36.5

= 0.03425.Volume =

= 30 $cm^3$

= 0.030 $dm^3$.Molarity =

= 0.03425 ÷ 0.030

= 1.14 mol/$dm^3$ (1.14M).

Question: 2.5 g KCl is dissolved in water to make a 100 $cm^3$ solution. Find the concentration (mol/$dm^3$).

Answer: Molar mass of KCl = 74.5 g.Moles =

= 2.5 ÷ 74.5

= 0.03356.Volume =

= 100 cm³

= 0.1 dm³.Concentration =

= 0.03356 ÷ 0.1

= 0.34 mol/dm³.

Question: A flask has a 0.5M NaOH solution. How much NaOH is in 1 dm³ of the solution?

Answer: Moles =

= 0.5 × 1

= 0.5. Molar mass of NaOH = 40 g, so mass = 0.5 × 40 = 20 g.

Question: How much 4M stock solution is needed to make 200 ml of 0.5M solution?

Answer: By M1V1 = M2V2, 4 × V1 = 0.5 × 200, so V1 = 100 ÷ 4 = 25 ml.

Question: A mineral-water bottle has a 100 cm³ solution containing 28 g calcium. Find its concentration in g/dm³.

Answer: Volume =

= 100 cm³

= 0.1 dm³.Concentration =

= 28 ÷ 0.1

= 280 g/dm³.

Question: 20 cm³ alcohol is mixed with 80 cm³ water. Find the (v/v) concentration of this solution.

Answer: Total volume =

= 20 + 80

= 100 cm³.(v/v)% =

= 20 ÷ 100 × 100

= 20% (v/v).

Question: How much NaOH is needed to prepare 400 cm³ of 0.3M solution?

Answer: Moles =

= 0.3 × 0.4

= 0.12. Molar mass of NaOH = 40 g, so mass = 0.12 × 40 = 4.8 g.

Question: In-text question: Why is a solution important in our life?

Answer: Most processes in our body and daily life happen in solution, such as food digestion, medicines working and cleaning.

Question: In-text question: Why is a solution called a mixture?

Answer: A solution is called a mixture because two or more substances are

uniformly mixed in it, without a chemical reaction.

Question: In-text question: What is an aqueous solution?

Answer: An aqueous solution is a solution in which the solvent is water.

Question: In-text question: Give examples of solute and solvent.

Answer: Salt in water: salt is the solute, water is the solvent. In air: nitrogen is the solvent, oxygen and carbon dioxide are solutes.

Question: In-text question: Air is a solution of gases like nitrogen, carbon dioxide and oxygen. Which gas is the solvent in air?

Answer: Nitrogen is the solvent in air, because it is present in the largest amount.

Question: In-text question: Beaker A has 20 mL water + 10 g sodium thiosulphate and beaker B has 20 g sodium thiosulphate. Which beaker has a saturated solution? [Note: the printed scenario is partly unclear.]

Answer: Beaker B has the saturated solution, because it has more solute and undissolved solid remains.

Question: In-text question: How can a supersaturated solution be prepared with this experiment?

Answer: The saturated solution is heated to dissolve more solute, then it is cooled slowly without disturbance, so a supersaturated solution forms.

Question: In-text question: Show the difference between saturated and unsaturated solution.

Answer: A saturated solution holds the maximum solute at that temperature and extra solute stays undissolved. An unsaturated solution can still dissolve more solute.

Question: In-text question: Which beaker has an unsaturated solution?

Answer: Beaker A has the unsaturated solution, because it has less solute and all of it dissolves.

Question: In-text question: How is an unsaturated solution prepared?

Answer: An unsaturated solution is prepared by dissolving less solute than the solubility limit.

Question: In-text question: In the lab there is 10 M $HNO_3$. How can 500 $cm^3$ of 0.1 M $HNO_3$ be prepared from it?

Answer: By $M_1V_1 = M_2V_2$, $10 \times V_1 = 0.1 \times 500$, so $V_1 = 5$ $cm^3$. Take 5 $cm^3$ of the 10 M acid and dilute it with water to 500 $cm^3$.

Question: In-text question: Define concentration of a solution.

Answer: Concentration of a solution is the amount of solute in a given amount of solution or solvent, e.g. percent or molarity.

Question: In-text question: Brass contains 20% zinc and 80% copper. Identify the solute and solvent and the type of solution.

Answer: Solute = zinc (smaller amount), solvent = copper (larger amount). Type of solution: solid-in-solid solution (alloy).

Question: In-text question: Differentiate between dilute and concentrated solution.

Answer: A dilute solution has little solute; a concentrated solution has much solute.

Question: In-text question: Which solution is more dilute: 2M or 3M?

Answer: The 2M solution is more dilute, because its molarity is lower.

Question: In-text question: A NaOH solution has concentration 1.2M. Find the mass of NaOH in 500 $cm^3$ in $g/dm^3$.

Answer: Mass in 1 $dm^3$ = $1.2 \times 40 = 48$ g, i.e. 48 $g/dm^3$. So mass in 500 $cm^3$ = 24 g.

Question: In-text question: Find the % (m/m) of a solution made by dissolving 10 g sugar in 140 g water.

Answer: Total mass =

= 10 + 140

= 150 g.% (m/m) =

= 10 ÷ 150 × 100

= 6.67%.

Question: In-text question: How is a 10% (m/m) sugar solution prepared? How much solvent is needed?

Answer: Dissolving 10 g sugar in 90 g water makes 100 g of 10% (m/m) solution. Solvent (water) needed = 90 g.

Question: In-text question: Explain the general rule "like dissolves like".

Answer: A polar solute dissolves in a polar solvent and a non-polar solute dissolves in a non-polar solvent, because their intermolecular forces are of the same kind.

Question: In-text question: Why does a solute dissolve in a solvent?

Answer: A solute dissolves because the solute-solvent attraction is stronger than the solute-solute and solvent-solvent forces.

Question: In-text question: If the solute-solvent attraction is stronger than the solute-solute attraction, can a solution form?

Answer: Yes, a solution can form, because the solvent separates the solute particles and surrounds them. [Note: the printed wording of the question is not fully clear.]

Question: In-text question: Why does benzene not dissolve in water?

Answer: Benzene is non-polar and water is polar, so benzene does not dissolve in water.

Question: In-text question: Why does a non-polar solute not dissolve in water? What is the reason?

Answer: The polar water molecules attract one another strongly by

hydrogen bonding, and a non-polar solute cannot break in between them.

Question: In-text question: How is a colloid different from a solution?

Answer: Colloid particles are larger than solution particles (1 to 1000 nm), they scatter light and look less homogeneous than a solution.

Question: In-text question: Which is a colloidal solution: starch solution or glucose solution?

Answer: Starch solution is a colloidal solution, whereas glucose solution is a true solution.

Question: In-text question: Why do the particles of a colloid not settle?

Answer: Colloid particles are very small and always in motion, so they do not settle at the bottom.

Question: In-text question: Why is a colloidal solution shaken well before use?

Answer: Shaking spreads the particles evenly again. [Note: the printed wording of the question is not fully clear.]

Question: In-text question: Give an example of a colloid with its use.

Answer: Milk is a colloid and is used as food. Fog is also a colloid.

Question: In-text question: Give an example of a colloid with its preparation.

Answer: To prepare a starch colloid, starch is mixed in hot water, which spreads its particles.

Question: In-text question: Give a reason why a colloid is a solution.

Answer: In a colloid the particles are spread evenly through the whole solution and do not settle, so it is called a solution.

Question: In-text question: How does a colloid scatter light?

Answer: Colloid particles scatter light rays in different directions, so the path of light becomes visible.

Question: In-text question: Describe a false (non-true) solution.

Answer: A false solution is one whose particles are larger than true-solution particles and which scatter light, such as a colloid or suspension.

Question: In-text question: When does a colloid show the Tyndall effect?

Answer: A colloid shows the Tyndall effect when a beam of light passes through it and the particles scatter it.

# برقي كيميا7باب : (Electro Chemistry)

Question: What is the Cu-Sn alloy called? (a) Brass (b) Bronze (c) Monel (d) Bell metal

Answer: Correct answer: (b) Bronze.

Question: Which of the following is not an electrolyte? (a) Graphite (b) Mercury (c) Steel (d) Water

Answer: Correct answer: (a) Graphite, because it conducts electricity through electrons, not ions.

Question: One Faraday equals how many coulombs? (a) 9.65C (b) 9650C (c) 96500C (d) 96C

Answer: Correct answer: (c) 96500C.

Question: Which of the following is not an electrolyte? (a) Aqueous HCl (b) Aqueous NaCl (c) KCl solution (d) Urea

Answer: Correct answer: (d) Urea, because it does not form ions in water.

Question: Which of the following is an oxidising agent? (a) Al (b) H2S (c) Cl2 (d) NaH

Answer: Correct answer: (c) Cl2, because it takes electrons and oxidises another substance.

Question: Which of the following is a reducing agent? (a) H2SO4 (b) HNO3 (c) Al (d) I2

Answer: Correct answer: (c) Al, because it gives electrons and reduces another substance.

Question: Which forms a weak electrolyte with water? (a) HCl (b) KOH (c) NaCl (d) CH3COOH

Answer: Correct answer: (d) CH3COOH (acetic acid), because it ionises only slightly in water.

Question: In a Daniel cell, what is used as the cathode? (a) Zn (b) Cu (c) Sn (d) Pb

Answer: Correct answer: (b) Cu; Zn is the anode.

Question: What is the equivalent weight of aluminium? (a) 9g (b) 27g (c) 54g (d) 1g

Answer: Correct answer: (a) 9g, because equivalent weight = atomic mass ÷ valency = 27 ÷ 3 = 9.

Question: Which of the following statements is correct? (a) Oxidation occurs at the cathode (b) Reduction and oxidation occur at the anode (c) Reduction occurs at the cathode and oxidation at the anode (d) The cathode gives electrons

Answer: Correct answer: (c) Reduction occurs at the cathode and oxidation at the anode.

Question: Define oxidation and reduction with examples.

Answer: Oxidation: losing electrons or gaining oxygen, e.g. $Zn \rightarrow Zn^{2+} + 2e^{-}$. Reduction: gaining electrons or losing oxygen, e.g. $Cl2 + 2e^{-} \rightarrow 2Cl^{-}$.

Question: Why do only ionic compounds conduct electricity in molten or aqueous state?

Answer: In ionic compounds, free ions can move in the molten or aqueous state and carry charge. In the solid state the ions are locked in place, so electricity cannot flow.

Question: What is an electrolytic cell or voltaic cell? Explain with a diagram.

Answer: A voltaic cell converts chemical energy into electrical energy, such as the Daniel cell. An electrolytic cell uses electrical energy to drive a chemical reaction, such as the electrolysis of water. Both have two electrodes (anode and cathode) and an electrolyte.

Question: Define oxidising and reducing agents with examples.

Answer: Oxidising agent: the substance that takes electrons and oxidises another, e.g. Cl2. Reducing agent: the substance that gives electrons and reduces another, e.g. Zn.

Question: Identify in this chemical equation: (i) oxidising agent (ii) reducing agent (iii) substance oxidised (iv) substance reduced. Equation: Zn + Cl2 → ZnCl2

Answer: Zn = reducing agent and substance oxidised. Cl2 = oxidising agent and substance reduced.

Question: Identify in this chemical equation: (i) oxidising agent (ii) reducing agent (iii) substance oxidised (iv) substance reduced. Equation: Br2 + H2S → 2HBr + S

Answer: H2S = reducing agent and substance oxidised. Br2 = oxidising agent and substance reduced.

Question: Identify in this chemical equation: (i) oxidising agent (ii) reducing agent (iii) substance oxidised (iv) substance reduced. Equation: 2Ca + O2 → 2CaO

Answer: Ca = reducing agent and substance oxidised. O2 = oxidising agent and substance reduced.

Question: Identify in this chemical equation: (i) oxidising agent (ii) reducing agent (iii) substance oxidised (iv) substance reduced. Equation: 2Li + S → Li2S

Answer: Li = reducing agent and substance oxidised. S = oxidising agent and substance reduced.

Question: Identify the alloy: Cu–Zn

Answer: Brass.

Question: Identify the alloy: Ni–Mg–Al–Cu

Answer: Duralumin.

Question: Identify the alloy: Sn–Zn–Cu

Answer: Bronze.

Question: Explain the dry cell with a diagram.

Answer: In a dry cell, the zinc cover is the anode, the carbon rod in the middle is the cathode, and NH4Cl paste between them is the electrolyte. Chemical reaction produces electricity. It is used in torches, radios and toys.

Question: What is a lead storage battery? How does it work?

Answer: A lead storage battery is a rechargeable battery with lead plates and sulphuric acid. On discharge, lead sulphate forms; on charging, lead and lead dioxide form again. It is used in vehicles.

Question: Explain the electroplating process steps.

Answer: (1) The object to be plated is cleaned. (2) It is made the cathode. (3) The plating metal is made the anode. (4) Both are dipped in electrolyte and current is passed, so the metal deposits on the cathode.

Question: What is an alloy? Explain its classification with examples.

Answer: An alloy is a homogeneous mixture of two or more metals. Examples: brass (Cu–Zn), bronze (Cu–Sn), steel (Fe–C), duralumin (Ni–Mg–Al–Cu).

Question: What is rusting? How can machines be protected from it?

Answer: Rusting is iron reacting with oxygen and water to form iron oxide. Protection: painting, greasing, galvanising (zinc coating) and plating.

Question: How can tin (Sn) and zinc (Zn) be electroplated on an iron surface?

Answer: The iron object is made the cathode and tin or zinc the anode; both are dipped in a solution of that metal's salt and current is passed, so tin or zinc deposits on the iron.

Question: How can silver be electroplated?

Answer: The object to be plated is made the cathode and silver the anode;

both are dipped in a solution of a silver salt (such as silver nitrate) and current is passed, so silver deposits on the object.

Question: State Faraday's first and second laws of electrolysis.

Answer: First law: the mass of substance deposited at an electrode is directly proportional to the quantity of electricity passed. Second law: when the same electricity is passed, the masses of different substances deposited are proportional to their equivalent weights.

Question: In-text question: Identify oxidising/reducing agents from: Al, Na, $H_2S$, $H_2SO_4$, $KMnO_3$, Zn.

Answer: Oxidising agents: $H_2SO_4$, $KMnO_3$. Reducing agents: Al, Na, $H_2S$, Zn.

Question: In-text question: Identify oxidising/reducing agents in: $H_2 + Cl_2 \rightarrow 2HCl$.

Answer: $H_2$ = reducing agent (oxidised). $Cl_2$ = oxidising agent (reduced).

Question: In-text question: Define electrolyte.

Answer: An electrolyte is a substance that conducts electricity in the molten or aqueous state through free ions.

Question: In-text question: What is a strong electrolyte?

Answer: A strong electrolyte is one that dissociates fully into ions in water, e.g. HCl, NaOH, KCl.

Question: In-text question: What is a non-electrolyte?

Answer: A non-electrolyte is a substance that does not produce ions and does not conduct electricity, e.g. glucose, urea, benzene.

Question: In-text question: Identify strong and weak electrolytes among: $HCl(aq)$, $KCl(aq)$, $NaOH(aq)$, $H_2S(aq)$, $CH_3COOH(aq)$, $NH_4OH(aq)$.

Answer: Strong: HCl, KCl, NaOH. Weak: $H_2S$, $CH_3COOH$, $NH_4OH$.

Question: In-text question: Draw a labelled diagram of the electrolytic cell for potassium chloride solution.

Answer: A beaker with KCl solution, two electrodes (anode and cathode), connected to a battery. $K^+$ moves to the cathode and $Cl^-$ to the anode.

Question: In-text question: Identify the cathode and anode in an electrolytic cell and explain oxidation, reduction and the movement of electrons.

Answer: Oxidation occurs at the anode (+), reduction at the cathode (−). Electrons flow from the anode to the cathode through the external circuit.

# باب8 ردعمل كيميائي : (Chemical Reactivity)

Question: Which metal belongs with the alkaline-earth metals? (a) B (b) Bi (c) Br (d) Ba

Answer: Correct answer: (d) Ba (barium), which is a Group IIA metal.

Question: Which of the following is a non-metal? (a) B (b) Be (c) Ba (d) Br

Answer: Correct answer: (d) Br (bromine), which is a halogen non-metal.

Question: Chlorine can be displaced by: (a) F (b) Br (c) I (d) At

Answer: Correct answer: (a) F (fluorine), because only fluorine is above chlorine and more reactive.

Question: Which is the strongest acid? (a) HF (b) HCl (c) HBr (d) HI

Answer: Correct answer: (d) HI, because the acidity of halogen acids increases in the order HF < HCl < HBr < HI.

Question: Which halogen is found in the solid state? (a) Fe (b) Cl2 (c) Br2 (d) I2

Answer: Correct answer: (d) I2 (iodine), which is a solid.

Question: The non-metals of Group ___ are found in the gaseous state. (a) VIA (b) VIIA (c) VIIIA (d) VIIIB

Answer: Correct answer: (c) VIIIA, i.e. the noble gases (He, Ne, Ar, Kr, Xe, Rn).

Question: Which of the following is a metalloid? (a) Br (b) Se (c) S (d) Sr

Answer: Correct answer: (b) Se. [Note: the chapter's own metalloid list (B, Si, Ge, As, Sb, Te, Po, At) does not include Se; Se is the closest choice and is classed as a metalloid in some references.]

Question: Which acts as an oxidising agent in a chemical reaction? (a) Be (b) Mg (c) Na (d) Cl

Answer: Correct answer: (d) Cl, because it takes electrons and oxidises

another substance.

Question: Which gas turns limewater milky? (a) O2 (b) NO2 (c) CO2 (d) N2

Answer: Correct answer: (c) CO2, because it combines with limewater to form calcium carbonate.

Question: Silver nitrate is used as the compound called Lunar Caustic. Which compound is it? (a) KNO3 (b) AgNO3 (c) NaOH (d) NaNO3

Answer: Correct answer: (b) AgNO3.

Question: Classify these elements as metal, non-metal, metalloid: Ge, Ba, P, K, Si, Sr, S, Sb, Ca, C.

Answer: Metalloids: Ge, Si, Sb. Metals: Ba, K, Sr, Ca. Non-metals: P, S, C.

Question: What are alkali metals? Write the names and symbols of alkali metals.

Answer: Alkali metals are the very reactive metals of Group IA. Names and symbols: lithium (Li), sodium (Na), potassium (K), rubidium (Rb), caesium (Cs), francium (Fr).

Question: What are alkaline-earth metals? Write their names and symbols.

Answer: Alkaline-earth metals are the metals of Group IIA. Names and symbols: beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba), radium (Ra).

Question: What are halogens? Write their names and symbols.

Answer: Halogens are the reactive non-metals of Group VIIA. Names and symbols: fluorine (F), chlorine (Cl), bromine (Br), iodine (I), astatine (At).

Question: Name any three metalloids with symbols.

Answer: Boron (B), silicon (Si), germanium (Ge).

Question: Describe with examples: 1. Cation 2. Anion.

Answer: 1. Cation: a positive ion, e.g. $Na^{+}$ and $Ca^{2+}$. 2. Anion: a negative ion, e.g. $Cl^{-}$ and $O^{2-}$.

Question: Why are alkali metals not found free in nature? Explain.

Answer: Alkali metals are very reactive and react quickly with air and water, so they cannot stay in the free state and occur in compounds.

Question: Explain the inertness of noble metals.

Answer: Noble metals such as gold, silver and platinum do not react easily with other substances, because they are chemically stable and are not affected by air, water and acids.

Question: Name some noble metals with symbols.

Answer: Gold (Au), silver (Ag), platinum (Pt), iridium (Ir), osmium (Os), ruthenium (Ru), rhodium (Rh), palladium (Pd).

Question: State the electronegativity property of non-metals.

Answer: Non-metals attract electrons strongly, so their electronegativity is high. Fluorine is the most electronegative element.

Question: Write a detailed note on the importance of silver.

Answer: Silver is a very good conductor of heat and electricity, and is used in jewellery, coins, utensils, photography, electronics and medicines. It is a noble metal and does not rust easily.

Question: State the properties and importance of gold.

Answer: Gold is a shiny, soft, malleable and rust-free metal. It is used in jewellery, coins, dentistry, electronics and space technology.

Question: Explain the test for chloride ($Cl^-$) and iodide ($I^-$) ions.

Answer: $Cl^-$ test: adding silver nitrate to the solution gives a white precipitate. $I^-$ test: adding silver nitrate gives a yellow precipitate.

Question: Write a note on the positive-charge property of metals.

Answer: Metals give electrons to form positive ions (cations), e.g. $Na \rightarrow Na^+ + e^-$. That is why metals conduct electricity and form ionic compounds.

Question: State the importance of magnesium in the periodic table.

Answer: Magnesium is a Group IIA alkaline-earth metal. It is important in

light strong alloys, fireworks, medicines and as part of chlorophyll in plants.

Question: State the importance of sodium in the periodic table.

Answer: Sodium is a Group IA alkali metal. It is used in salt, soda, soap, glass and chemical industries. It also plays an important role in the body.

Question: Arrange the halogens' acids in order of increasing acidity: HF, HI, HCl, HBr.

Answer: HF < HCl < HBr < HI, i.e. HF is the weakest and HI is the strongest.

Question: Explain electronegativity and metallic properties of non-metals.

Answer: Non-metals have high electronegativity, so they take electrons. They lack metallic properties: they are not shiny, not conductors of electricity and are brittle.

Question: Differentiate between soft metals and hard metals (sodium vs iron).

Answer: Soft metal (sodium): can be cut with a knife, Mohs hardness 0.5, melting point 98°C, density 0.971 g/cm³. Hard metal (iron): Mohs hardness 4.5, melting point 1535°C, density 7.87 g/cm³.

Question: Discuss the reactivity of halogens.

Answer: Halogens are very reactive non-metals, because their outer shell has 7 electrons and taking one electron completes the octet. Reactivity decreases from top to bottom: F > Cl > Br > I.

Question: In-text question: Which metals are found in the liquid state?

Answer: Mercury (Hg) and gallium (Ga) are found in the liquid state.

Question: In-text question: Identify the alkaline-earth metal elements from: Sr, Sb, Si, Se, B, Ba, Br, Bi.

Answer: Ba and Sr are the alkaline-earth metals.

Question: In-text question: Write a few properties of metals.

Answer: Metals are lustrous, malleable and ductile, good conductors of heat and electricity, sonorous, have high melting points, and form cations.

Question: In-text question: Write a few properties of non-metals.

Answer: Non-metals are dull, brittle, poor conductors of electricity, have low melting and boiling points, and form anions.

Question: In-text question: Show the uses of washing soda, bleach and sodium nitrate.

Answer: Washing soda (Na2CO3): cleaning, water softening, making glass and soap. Bleach (Ca(OCl)2): disinfectant, whitening. Sodium nitrate (NaNO3): fertiliser, food preservation, glass.

Question: In-text question: Show the uses of magnesium.

Answer: Magnesium is used in alloys, fireworks/flares, medicine (antacid), chlorophyll in plants and sacrificial protection.

Question: In-text question: Show the uses of lime, plaster/gypsum and calcium tungstate.

Answer: Lime: cement, mortar, whitewash, steel. Gypsum/plaster: plaster of Paris, wallboard, cement. Calcium tungstate: fluorescent lamps, X-ray screens.

Question: In-text question: Describe soft metals with examples.

Answer: Soft metals are those that can be cut with a knife and have low hardness and melting points, such as Na, K, Rb, Cs.

Question: In-text question: Describe hard metals with examples.

Answer: Hard metals are those with high hardness and melting points and strong metallic bonding, such as Ni, Fe, W.

Question: In-text question: Show the values of melting point, boiling point, density and Mohs hardness of sodium and iron.

Answer: Na: melting point 98°C, boiling point 890°C, density 0.971 g/cm³, Mohs 0.5. Fe: melting point 1535°C, boiling point 2450°C, density 7.87

g/cm³, Mohs 4.5.

Question: In-text question: Write the names of noble metals with symbols.

Answer: Gold (Au), silver (Ag), platinum (Pt), iridium (Ir), osmium (Os), ruthenium (Ru), rhodium (Rh), palladium (Pd).

Question: In-text question: Why is an astronaut's helmet visor coated with gold?

Answer: Gold reflects infrared/solar radiation and protects the astronaut from heat.

Question: In-text question: Why is a gold layer applied to mirror glass?

Answer: Gold is highly reflective, does not tarnish and protects from heat. [Note: the printed wording of the question is not fully clear.]

Question: In-text question: What is the use of gold in dentistry?

Answer: Gold is inert, non-corrosive and biocompatible, so it is used in fillings and crowns.

Question: In-text question: What is platinum used for as a catalyst?

Answer: Platinum speeds up reactions without being consumed, e.g. in catalytic converters and the Ostwald process.

Question: In-text question: State the properties of non-metals.

Answer: Non-metals are dull, brittle, poor conductors of electricity, have low melting and boiling points, gain electrons to form anions, and form acidic oxides.

Question: In-text question: Identify the elements of Group VIIIA from: N, Na, Ni, Ne, Ar, At, He.

Answer: Ne, Ar, He are the Group VIIIA elements.

Question: In-text question: Write the names and symbols of the Group VA non-metals.

Answer: Nitrogen (N), phosphorus (P), arsenic (As), antimony (Sb), bismuth (Bi).

Question: In-text question: Which group has only non-metals in the gaseous state?

Answer: Group VIIIA (noble gases) has only non-metals in the gaseous state.